THE TOTAL COMPLEX OF A TENSOR PRODUCT OF K-FLAT COMPLEXES IS K-FLAT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument: the displayed associativity isomorphism is the computational content and the definition of K-flatness supplies the conclusion.

Editorial note. The counted spans contain no label reference and no bibliography citation, so the card carries no navigation stub and no bibliography excerpt.

Editorial note. Cross-chapter restatement: the same label occurs in chapter cohomology and in chapter more-algebra. Those statements are the special cases of this ringed-site statement, so per the lane-s4 rule only this more general occurrence is counted here.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- K-flat complex of O-modules on a ringed site. Same chapter, the definition of K-flatness: the total tensor product with every acyclic complex of O-modules is acyclic. Supplies both the hypothesis on K^bullet and L^bullet and the property to be verified for their total tensor product.
- Total complex of a double complex. Same chapter, the Tot construction and its compatibility with tensor products of complexes. Supplies the complex whose acyclicity is tested, and the associativity isomorphism displayed in the proof.
- Associativity of the total tensor product up to canonical isomorphism. Same chapter, the canonical isomorphism Tot(M^bullet tensor Tot(K^bullet tensor L^bullet)) = Tot(Tot(M^bullet tensor K^bullet) tensor L^bullet). Supplies the displayed isomorphism, which is the entire computational content of the proof.

Lemma 0.1. *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. If $\mathcal{K}^\bullet$, $\mathcal{L}^\bullet$ are K-flat complexes of $\mathcal{O}$-modules, then $Tot(\mathcal{K}^\bullet \otimes_{\mathcal{O}} \mathcal{L}^\bullet)$ is a K-flat complex of $\mathcal{O}$-modules.*

Proof. Follows from the isomorphism

$$\mathrm{Tot}(\mathcal{M}^\bullet \otimes_{\mathcal{O}} \mathrm{Tot}(\mathcal{K}^\bullet \otimes_{\mathcal{O}} \mathcal{L}^\bullet)) = \mathrm{Tot}(\mathrm{Tot}(\mathcal{M}^\bullet \otimes_{\mathcal{O}} \mathcal{K}^\bullet) \otimes_{\mathcal{O}} \mathcal{L}^\bullet)$$

and the definition. □

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